

MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 15

1. *Find the eigenvectors and eigenvalues of the following matrices:*

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \begin{bmatrix} 4 & 2 & 2 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}.$$

(Challenge question)*: For each of the above matrices, are the eigenvectors corresponding to different eigenvalues orthogonal?*

2. *Suppose A is an $n \times n$ matrix with complex coefficients (i.e., coefficients in $\mathbb{C}$). If $A^2 = -I$, what are the possible eigenvalues of A? If A has real coefficients (i.e., coefficients in $\mathbb{R}$), show that n* ***must*** *be even. Give an example of a matrix A such that $A^2 = -I$ and A has real coefficients.*